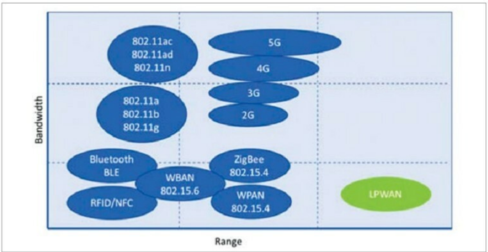
their vehicles, deployed over CAN Bus protocol. Data is transmitted from the vehicles to the IoT cloud through a mobile network."

Through a Web interface, the company can track their vehicles and gather more real-time intricate details such as rash driving by the drivers, impacts faced by the vehicle and cars usage approval details. Most importantly, the company can turn off any vehicle from their central control if it is under improper use. Detailed reports enable the organisation to optimise their services and greatly reduce financial losses.

—Paromik Chakraborty, technical journalist, EFY



LPWAN occupies the lowest bandwidth and provides the highest range of coverage, best fit for small data transmission (Credit: SenRa Tech Pvt Ltd)

Is India Reaping The Benefits Of Low Power Communication Yet?

Transceivers used in LPWAN remain off throughout and wake up only when a data signal needs to be transmitted or received

Low power communication technologies are reshaping the way data is transmitted today. While mobile and Wi-Fi services remain the primary mode of communication, smartcities and IoT-powered applications can leverage better from low-powered wide area networks (LPWANs).

Kishore S. Nambiar, head - business development and innovation, Unlimit IoT Pvt Ltd, says about India's adoption of LPWAN, "In the past one year, we have witnessed many successful implementations, especially in the areas of smart communities and advanced metering infrastructure (AMI). Smart City initiatives is further fueling its large-scale adoption in surveillance, smart public infrastructure, intelligent transportation systems, logistics and more."

How LPWAN caters to network needs

Distributed sensors in smartcity applications must run wirelessly with minimal supervision. Ali Hosseini, chief executive officer and director, SenRa Tech Pvt Ltd, says, "LPWAN solutions provide city-wide solutions with minimal impact on financial commitments, infrastructure, maintenance and management of deployed solutions. LPWAN technology is suited for solutions that only require devices to send small data over a wide area while maintaining battery life over many years."

Transceivers used in LPWAN remain off throughout and wake up only when a data signal needs to be transmitted or received. This is in contrast to mobile technology, where the transceiver remains active all the time. LPWAN, as a result, is able to conserve the power stored in the coin-cell battery that runs it. These devices can run for up to ten years without any need of changing the cell.

If LPWAN is compared to networks like Wi-Fi, the latter will have a shorter distance of coverage. LPWAN technologies like LoRa cover a range of 5km to 15km. Long distance Wi-Fi, in comparison, covers up to a maximum of 500 metres.

LPWAN technologies use an unlicensed radio-frequency spectrum, at 865MHz to 867MHz, making adoption by solution providers easier and hassle-free. Moreover, the networks are securely built.

Case Study: Maharashtra greatly improves water distribution using LoRaWAN

A city in Maharashtra has applied LPWAN solutions to solve stringent water supply problems. Inconsistent water distribution throughout the city was a longstanding challenge. Due to lack of infrastructure, further issues

WHY LoRaWAN WAS SELECTED FOR THE PROJECT

Smart city requirements	LoRaWAN	SigFox	Ingenu	LTE NB-IoT	Cellular	LAN (Wi-Fi/BLE)
Standards	Yes, Open (Lora Alliance)	No, proprietary	No, proprietary	Yes	Yes	Yes
Built-in security	Yes	No	Yes	TBD	No	Varies
Bi-directional communication	Yes	Limited	Yes	Yes	Yes	Yes
Interference	Unlicensed 865MHz	Unlicensed 865MHz	Unlicensed 2.4GHz	Licensed regionalised	Licensed regionalised	Unlicensed 2.4GHz

Source: Ali Hosseini, SenRa